

sorting speed in MB/s

5000

4000

3000

2000

1000

0

Nvidia Grace CPU Superchip, Neoverse-V2, 144 cores, 3.4 GHz
470 GiB RAM

—○— parallel, 4 pinned threads
—×— radsort, 1 thread
—+— generic, 1 thread

—□— bitmanip, 1 thread
—△— swc, 1 thread

$\sigma = 256$ (one byte)
 $n_t = 4$ (key); 4 bytes val.
 $b = 512$

cacheline ->

L1 cache ->

L2 cache ->

L3 cache ->

RAM of 1st NUMA node

$2^5 = 32$ B

$2^{10} = 1$ KiB

$2^{15} = 32$ KiB

$2^{20} = 1$ MiB

$2^{25} = 32$ MiB

$2^{30} = 1$ GiB

$2^{35} = 32$ GiB

array length [bytes]

△ larger is better

